Study Design Trade-offs

We describe key design decisions in our study and discuss the trade-offs based on the guidelines outlined by Robillard et al. [2].

Decision #1: Recruiting Technical Writers

We decided to focus on technical writers for the interview study, despite the documentation review process having several practitioner roles like developers and program managers.

Alternatives. We could have sampled across all practitioner roles involved or conducted separate studies for each group.

Rationale. Technical writers oversee documentation processes in their organizations, which gives them them visibility into how review practices work. Since documentation is their primary responsibility, they can speak in detail about review processes and quality issues. Other practitioners typically engage with documentation review as one component among other responsibilities. Due to these two reasons, technical writers can be the ideal informants to understand the documentation review process.

Considerations. Since our study is the first to investigate the documentation review process, we did not know the full range of roles involved in documentation review. We needed to understand the landscape first before involving other roles. Also, existing research has already looked at developers' perspectives on documentation [1, 3], while technical writers' perspectives has not been studied. Finally, technical writers have active communities that made it feasible for the researchers to perform target recruiting.

Implications. This means we capture documentation review from the perspective of primarily one practitioner role. This indeed provides understanding of the overall process, but it might miss specific challenges, motivations, or quality criteria from other practitioner roles. Future work can examine perspectives of other roles involves in the documentation review process.

Decision #2: Recruitment Pool

We recruited our participants from professional communities like Write The Docs and LinkedIn, and in addition requested for published documentation which resulted in recruiting experienced practitioners rather than getting participants broadly across experience levels.

Alternatives. We could have recruited from general software development communities or included practitioners with less documentation experience.

Considerations. Having initially decided to focus on technical writers, we scoped our recruitment to this practitioner role and maintained this focus throughout the study, and therefore recruiting from these groups meant that we had high quality participants. We required participants to provide us with documentation, in order to verify that they had the necessary experience and could function as informants and provide us with high quality data.

Rationale. While our recruitment attracted experienced technical writers, this indeed was beneficial for our goals because experienced practitioners have exposure to the review processes over the years and deeper insights into challenges. This was the focus of our study, and not to compare across the experience levels and how practitioners with different experience levels function within the review process.

Implications. While we still had participants from startup environments and organizations where review may be informal and minimal, we still underrepresent these contexts. The companies that invest in hiring for a dedicated technical writing role might be prioritizing the documentation quality already, and therefore might have systematic review process than other companies where the review is informal. Potentially due to recruiting from these communities, our pool of participants has experienced technical writers, which means that the challenges might be different for novice technical writers. Additionally, we do not speak to how review practices differ based on practitioner experience level.

An unintended consequence of these recruitment choices we ended up with participants having experience with different subsets of review stages rather than all participants experiencing all stages. Individual participants typically experienced 3-4 of the five review stages we identified in our analysis. This demonstrates that different organizations prioritize different aspects of the review as discussed in Section 3.2.1 (Overview of Documentation Review Process). After discovering this variation, we could have recruited to make sure that the participants practiced all review stages, but this would need modifying the screening criteria. However, this ended up strengthening the study since it provides insights into how different organizations approach the review process and their key considerations.

References

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